MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 4

1. *Find the inverses of the following matrices or show that the matrix is not invertible:*

(1) $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

(2) $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

(3) $\begin{bmatrix} 2 & 1 & 0 & -1 \\ 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

(4) $\begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix}$.

2. *Determine whether each of the following matrices are invertible. Justify your answer. It is not necessary to find the inverse if it exists (but you can if it helps you determine if it is invertible).*

(1) $\begin{bmatrix} 5 & 7 \\ -3 & -6 \end{bmatrix}$

(2) $\begin{bmatrix} 5 & 0 & 0 \\ -3 & -7 & 0 \\ 8 & 5 & -1 \end{bmatrix}$

(3) $\begin{bmatrix} 0 & 3 & -5 \\ 1 & 0 & 2 \\ -4 & -9 & 7 \end{bmatrix}$

(4) $\begin{bmatrix} -1 & 3 & 0 & 1 \\ 3 & 5 & 8 & -3 \\ -2 & -6 & 3 & 2 \\ 0 & -1 & 2 & 1 \end{bmatrix}$

3. *Answer the following true-false questions. Justify your answers! A and B are arbitrary $n \times n$ square matrices.*

(1) *If A is invertible then $A\mathbf{x} = \mathbf{0}$ has exactly one solution, $\mathbf{x} = \mathbf{0}$.*
(2) *If A is invertible, then AB is also invertible.*
(3) *If A and B are invertible, then $A + B$ is also invertible.*
(4) *If A is invertible, then the reduced echelon form of A is equal to I.*
(5) *If A is invertible and has an LU decomposition $A = LU$, then both L and U are invertible.*
(6) *If A is not invertible and has an LU decomposition $A = LU$, then neither L nor U are invertible.*

(7) *If A has an LU factorization, then this factorization is unique.*

4. *Let*

$$A = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{bmatrix}.$$

Such a matrix called ***tridiagonal matrix****. Band matrices often appear in applications (see the next exercise) and they are one reason LU-decomposition is so important.*

(1) *Find the LU-decomposition of A.*
(2) *Determine A^{-1}.*

Given a vector $\boldsymbol{b}$ in $\mathbb{R}^5$, suppose we want to solve $A\boldsymbol{x} = \boldsymbol{b}$. In many applications, we probably don't want to do this for a single $\boldsymbol{b}$, but for many (maybe a million) different vectors $\boldsymbol{b}$. One way for finding $\boldsymbol{x}$ is to solve the following two linear systems which arise from the LU-decomposition of A:

$$L\boldsymbol{c} = \boldsymbol{b}, \text{ and } U\boldsymbol{x} = \boldsymbol{c}.$$

Another way is to calculate A^{-1}, and then we would get $\boldsymbol{u}$ by simply multiplying $\boldsymbol{b}$ by A^{-1}. Which way is more efficient?

(3) *Given the LU-decomposition of A, count the numbers of operations (addition and multiplication of real numbers) needed to find $\boldsymbol{x}$ using first method?*
(4) *Given the inverse of A, count the number of operations needed to calculate $A^{-1}\boldsymbol{b}$?*
(5) *Suppose A is not a 5×5 band matrix, but a 1000×1000 band matrix? Which of the two methods would you use?*